D-G-00675a(GSA-FSS)
June 6, 1957
Revision of
Int. Fed. Spec. D-G-00675(GSA-FSS)
October 20, 1954

INTERIM FEDERAL SPECIFICATION

D-G-00675a(GSA-FSS)

GRENADES; HAND, GAS

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington 25, D. C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification covers tear and neaseating gas hand grenades for use by law enforcement officers, when necessary, for mob dispersion, evacuating dangerous and insame persons, etc., from buildings and other barricaded locations; also for the protection of law enforcement personnel. The grenades specified herein include container, firing mechanism, fuse, and gas-agent charge.

1.1.1 Federal specification coverage.- Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and loading composition.- The hand grenades covered by this specification shall be of the following types, classes, and loading compositions, as specified in the invitation for bid, contract, or order (see 6.1).

Type I.- Blast dispersion (for mob dispersal or barricade drive-out).
  Class A.- Single blast (see 3.7.1 and table I).
    Composition:
      No. 1.- CN, or CN-EBA, tear gas (see 3.1.2.1).
      No. 2.- DM nauseating gas (see 3.1.2.2).
      No. 3.- CN-DM, or CN-DA-EBA, tear and nauseating gas (see 3.1.2.3).
  Class B.- Triple blast (see 3.7.2 and table I).
    Composition:
      No. 1.- CN, or CN-EBA, tear gas.
      No. 3.- CN-DM, or CN-DA-EBA, tear and nauseating gas.

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Type II.- Combustion or continuous discharge.
Class A.- Single unit standard candle (for mob dispersal). (see 3.8.1 and table I).
Composition:
No. 1.- CN, or CN-EBA, tear gas.
No. 2.- DM nauseating gas.
No. 3.- CN-DM, or CN-DA-EBA, tear and nauseating gas.

Class B.- Single unit flameless candle (for barricade drive-out) (see 3.8.2 and table I).
Composition:
No. 1.- CN tear gas.

Class C.- Triple unit standard candle (for mob dispersal) (3.8.2 and table I).
Composition:
No. 1.- CN, or CN-EBA, tear gas.
No. 2.- DM nauseating gas.
No. 3.- CN-DM, or CN-DA-EBA, tear and nauseating gas.

2. APPLICABLE SPECIFICATION AND OTHER PUBLICATIONS

2.1 Specification.- The following specifications of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specification:

O-D-390 - Diphenylchloroarsine (DA)

Military Specifications:

MIL-C-10338 - Chloracetophenone (CN)
MIL-D-11772 - Diphenylaminechlorarsine (DM)

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under "General Information" in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

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(Copies of the Military Specifications referenced above, required by the contractor in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following publications, of the issue in effect on the date of invitation for bids, form a part of this specification:

Governmental:

Interstate Commerce Commission Regulation for Transportation of Explosives and Other Dangerous Articles, Etc.

(Copies of Interstate Commerce Commission Regulation for Transportation of Explosives and Other Dangerous Articles, Etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.).

Nongovernmental:

"War Gases" 1942 Edition, by Morris B. Jacobs

(Copies of the publication "War Gases" may be procured from the Interscience Publishing Company, 250 Fifth Avenue, New York 1, N.Y.)

3. REQUIREMENTS.

3.1 Materials.- All materials used in the construction of the hand grenades covered by this specification shall be of a quality adequately suitable for the purpose intended. Material shall be free from all defects and imperfections that might affect the serviceability or safety characteristics of the finished product.

3.1.1 Components of tear and nauseating gases.- For the purpose of brevity, the components of tear and nauseating gases will be referred to in this specification by their commonly designated symbols in lieu of their chemical names and shall conform to the requirements of the following sub-paragraphs.

3.1.1.1 Chloracetophenone ($C_6H_5COCH_2Cl$).- Chloracetophenone commonly designated as CN and known as tear gas shall conform to the physical and chemical requirements of Military Specification MIL-C-10338. (See 2.1).

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3.1.1.2 Diphenylaminechlorarsine $(C_6H_4)_2NHAsCl$.- Diphenylaminechlorasine commonly designated as DM and known as nauseating gas shall conform to the physical and chemical requirements of Military Specification MIL-D-11772 (see 2.1).

3.1.1.3 Diphenylchloroarsine $(C_6H_5)_2AsCl$.- Diphenylchloroarsine commonly designated as DA and known as nauseating gas shall conform to the physical and chemical requirements of Federal Specification O-D-390 (see 2.1).

3.1.1.4 Ethyl bromoacetate ($CH_2BrCOOC_2H_5$).- Ethyl bromoacetate commonly designated as EBA and known as tear gas shall conform to the chemical and physical properties whcih are given in the applicable sections of the publication "War Gases", 1942 edition (see 2.2).

3.1.2 Compositions of tear and tear and nauseating gases.- Unless otherwise specified, at the option of the contractor, composition No. 1 (tear gas) and composition No. 3 (tear and nauseating gases), may be furnished in either solid or liquid form (see 6.2).

3.1.2.1 Composition No. 1, DN, or CN-EBA, tear gas.- Composition No. 1, tear gas shall be either of two forms as follows:

CN as a solid, or CN-EBA as a liquid.

The CN shall conform to the requirements as specified in 3.1.1.1.
The CN-EBA shall be a combination of CN as specified in 3.1.1.1, dissolved in not more than 75 percent of EBA which shall conform to the requirements specified in 3.1.1.4.

3.1.2.2 Composition No. 2 DM, nauseating gas.- Composition No. 2, nauseating gas shall conform to the requirements specified in 3.1.1.2.

3.1.2.3 Composition No. 3, CN-DM, or CN-DA-EBA, tear and nauseating gas.- Composition No. 3, tear and nauseating gas shall be either of two forms as follows:

CN-DM as a solid, or CN-DA-EBA as a liquid.

The CN-DM composition shall be a combination of CN conforming to the requirements specified in 3.1.1.1, and DM conforming to the requirements specified in 3.1.1.2.
The CN-DA-EBA composition shall be a combination of CN conforming to the requirements specified in 3.1.1.1, and DA conforming to the requirements specified in 3.1.1.3. The components CN and DA shall be dissolved in not more than 75 percent of EBA which shall conform to the requirements specified in 3.1.1.4.

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3.1.3 Grenade container.- The grenade container shall be metal of adequate strength for the intended purpose.

3.2 Standard products.- The hand grenades shall be essentially the standard product of the manufacturer. The component parts of the grenade need not be the product of the same manufacturer.

3.3 Design.

3.3.1 Grenade outer container.- The outer container shall be metal of adequate strength to protect the grenade from damage in handling and so designed as to liberate the gas without fragmentation of the grenade. The metal shall be resistant to corrosion by its composition or by a protective plating.

3.3.2 Grenade firing mechanism.- Each hand grenade shall be supplied with a firing mechanism constructed of noncorrosion metal. The firing mechanism shall be of such construction as to assure positive firing action and also when operated in accordance with the precautionary and operating instructions printed on the container of the grenade, to assure absolute protection against discharge of the grenade until the safety pin or other similar device is manually removed from the firing mechanism and the grenade released from the hand in the act of throwing. The involvement of the gas shall start within 1 to 2-1/4 seconds after the grenade is thrown.

3.3.3 Dispersal of tear and nauseating gases.- The gases shall be completely liberated from the grenade within 20 to 35 seconds.

3.4 Instruction and caution labeling.- Each hand granade shall be conspicuously and permanently marked with legible and concise caution and operating instructions.

3.5 Identification coating and marking.- Each hand grenade shall be identified by means of a color code to indicate the specific type of gas agent, or agents with which the grenade is loaded. The color code identification marking shall be as follows:

(a) For CN, or CN-EBA (tear gas) loaded grenades, the indicating color shall be RED.
(b) For DM, or DA-EBA (nauseating gas) loaded grenades, the indicating color shall be GREEN.
(c) For CN-DM or CN-DA-EBA-(tear and nauseating gas) loaded grenades, the indicating color shall be RED AND GREEN.

In addition to the color code marking, the hand grenades shall also be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined; and also be marked as to type, delayed firing time, etc., and the expiration date of the manufacturer's guarantee. They may be further identified in accordance with any of the manufacturer's data.

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3.6 Storage.- Under normal storage conditions (cool and dry) the loaded hand grenades shall give off no obnoxious vapors or odors and they shall remain fully effective for a perion of at least three years from date of manufacture.

3.7 Type I, blast dispersion grenade.- The type I, blast dispersion grenade shall be constructed so that when discharged it shall release either a single or series of blasts of gas without expelling fragments of the grenade.

3.7.1 Type I, class A, grenade.- The type I, class A, grenade shall be constructed so that when discharged it will release a single blast of gas. The grenade shall be loaded with either composition No. 1, CN, or CN-EBA (tear gas), composition No. 2, DM (nauseating gas), or composition No. 3, CN and DM or CN-DA-EBA (tear and nauseating gas), as specified in the invitation for bids, contract or order (see 6.1(b)). The dimensions, weight, and gas-agent load of the grenade shall conform to the applicable requirements of table I.

3.7.2 Type I, class B, grenade.- The type I, class B, grenade shall be constructed so that when discharged it will release three separate blasts of gas. The first blast to occur approximately 1-3/4 seconds after the grenade is thrown. The second and third blasts to occur at intervals of approximately one second. On each blast the grenade shall make a jump of not less than four feet. The grenade shall be loaded with either composition No. 1 (CN, or CN-EBA) (tear gas), or composition No. 3, CN-DA-EBA (tear and nauseating gas), as specified in the invitation for bids, contract or order (see 6.1(b)). The dimensions, weight, and gas-agent load of the grenade shall conform to the applicable requirements of table I.

3.8 Type II, combustion or continuous discharge grenades.- Type II, combustion or continuous discharge grenades, shall be constructed so as to consist of a single or series of individual grenades.

3.8.1 Type II, class A, grenade.- The type II, class A, grenade shall be single unit type constructed to generate sufficient heat to cause severe burns to an individual picking it up barehanded while the gas content of the grenade is being liberated. The grenade shall be loaded with either composition No. 1, CN (tear gas), composition No. 2, DM (nauseating gas), or composition No. 3, CN and DM (tear and nauseating gas) as specified in the invitation to bid, contract, or order (see 6.1(b)). The dimensions, weight, and gas-agent load of the grenade shall conform to applicable requirements of table I.

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3.8.2 Type II, class B, grenade.- The type II, class B, grenade shall be a single unit grenade constructed so as to generate sufficient heat to cause severe burns to an individual picking it up barehanded while the gas content is being liberated; however, not hot enough to start a fire when it is thrown into a quantity of dry and loose excelsior or shreded paper and left therein until completely discharged and cold. The loose and dry excelsior or shredded paper shall be of sufficient quantity to allow the grenade to sink into it so as to be completely surrounded with the excelsior or paper. The grenade shall be loaded with composition No. 1, CN (tear gas). The dimensions, weight and gas-agent load of the grenade shall conform to the applicable requirements of table I.

3.8.3 Type II, class C, grenade.- The type II, class C, grenade shall be constructed so as to consist of three sections or units which will separate two seconds after throwing, each section or unit functioning as an individual grenade. The grenade sections or units shall generate sufficient heat to cause severe burns to an individual picking them up barehanded while the gas content is being liberated. The grenade shall be loaded with either composition No. 1 CN (tear gas), composition No. 2, DM (nauseating gas), or composition No. 3, CN-DM (tear and nauseating gas), as specified in the invitation for bid, contract, or order (see 6.1). The dimensions, weight and gas-agent load of the grenade shall conform to the applicable requirements of table I.

3.9 Certification.- The manufacturer shall submit with his bid evidence that the grenades which he proposes to supply meet all the requirements of this specification, the requirements of his product as advertised, and the performance claims as set forth in the questionnaire of this specification.

3.10 Workmanship.- The grenades shall be free from defects resulting from inferior workmanship which may affect their appearance or impair their serviceability.

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Table I.- Dimensions, weights, and loading requirements for hand grenades.

Type	Class	Hand grenade dimensions and weights: Dimensions: Height, Inches: Min.	Height, Inches: Max.	Diameter, Inches: Min.	Diameter, Inches: Max.	Weight, Ounces: Min.	Weight, Ounces: Max.	Hand grenade loading compositions: Composition C No. 1 1/ CN or CN-EBA Tear gas loaded grenade CN grams: Min.	Composition C No. 1 1/ CN or CN-EBA Tear gas loaded grenade CN grams: Max.	Composition No. 2 DM Nauseating gas-loaded grenade DM grams: Min.	Composition No. 2 DM Nauseating gas-loaded grenade DM grams: Max.	Composition No. 3 CN-DM or CN-DA-EBA Tear and nauseating gas-loaded grenades: CN grams: Min.	CN grams: Max.	DM or DA grams: Min.	DM or DA grams: Max.
I	A	5-3/8	5-3/4	2-3/8	2-3/4	14	20	70	132	130	---	58	65	12	65
	B	6	6-7/8	2-5/8	3	25	35	100	---	---	---	85	---	15	---
II	A	4-7/8	5-3/4	2-3/8	2-3/4	17	22	90	123	138	---	45	66	45	65
	B	4-7/8	5-3/4	2-5/8	3	12	22	30	---	---	---	---	---	---	---
	C	4-7/8	5-3/4	2-5/8	3	17	22	78	---	120	---	58	---	59	---

NOTE: 1/ Compositions No. 1 and 3 shall contain not more than 75 percent EBA (see 3.1.2.1 and 3.1.2.3).

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4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 General.- The hand grenades and all the material entering into the manufacture thereof, shall be subject to inspection by authorized Government inspectors. When inspection is conducted at the contractor's plant, tests shall be conducted by the contractor under the supervision of an authorized Government inspector. The Government inspector may, at his option, conduct any and all of the tests. The place of inspection and acceptance shall be as specified in the invitation for bids, contract, or order (see 6.1).

4.2 Sampling.

4.2.1 Inspection lot.- For the purpose of inspection, at least five sample grenades (in no instance less than two percent of the quantity under inspection) of each type, class, and composition, shall be taken at random from those under inspection and submitted to inspection as specified in 4.3.

4.2.2 Test lot.- On orders of 500 or more grenades of one type, class, and composition, five sample grenades, or one percent of the order, shall be selected at random as samples for functional discharge testing.

4.3 Inspection.- The sample grenades selected in accordance with 4.2.1 shall be given a general inspection for material, workmanship, dimensions, weight, instruction and caution labeling, identification marking, packaging, packing, and general compliance with the requirements of this specification.

4.3.1 Acceptance and rejection.- Failure in more than two instances to meet the requirements of 4.3 shall be basis for rejection of the entire lot representative of the inspection lot.

4.4 Performance test.- The sample grenades selected in accordance with 4.2.2 shall be discharged for the purpose of determining conformance with the requirements of this specification, and performance in accordance with the manufacturer's certification, and standard performance claims as set forth by the manufacturer in the questionnaire, page 12, of this specification, and the manufacturer's printed literature and specification sheets.

4.4.1 Acceptance and rejection.- If more than one of the sample grenades selected in accordance with 4.2.2, and tested in accordance with 4.4 fails to function properly, i.e., discharge its gas content in accordance with the requirements of this specification, manufacturer's certification and standard performance claims, the manufacturer shall be required to submit five additional grenades selected from the same lot, in accordance with 4.2.2, for retest.

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4.4.2 Retest.- Failure of more than one grenade of this retest group to function in accordance with 4.4, shall be cause for rejection of the entire lot and the manufacturer shall then be required to submit an entirely new lot for test, with same sampling and test procedure herein specified, to be followed before final acceptance.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Unless otherwise specified, granades shall be contained in packages clearly marked with the manufacturer's name, nature of contents, and other applicable data such as: type, class, composition, etc. The packages shall be placed in cases which comply with Interstate Commerce Commission Regulations (see 2.2 and 6.1).

5.2 Packing.- The subject commodity shall be packed in accordance with Interstate Commerce Regulations for the transportation of explosives and other dangerous articles, etc. (see 2.2 and 6.1).

5.3 Marking.- Unless otherwise specified, shipping containers shall be clearly marked on the outside with the name of the commodity, the name of the contractor, and the number of the contract or order and the full name and address of the consignee; also, any other markings necessary to comply with Interstate Commerce Commission Regulations (see 6.1).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Type, number and date of this specification.
(b) Type, class and loading composition required (see 1.2.1, 3.7.1, 3.7.2, 3.8.1, 3.8.2, 3.8.3 and table I).
(c) Place of inspection and/or test (see 4.1).
(d) Packaging, packing, and marking, if different than specified (see 5.1, 5.2, and 5.3).

6.2 Information regarding grenade loading compositions.- Grenades covered by this specification are of visible and invisible gas-cloud compositions as follows:

Grenades classified as CN, DM, and CN-DM compositions are known as solid or dry-loaded grenades and when discharged they liberate a visible gas-cloud.

Grenades classified as CN-EBA, DA-EBA, and CN-DA-EBA are known as liquid-loaded grenades and when discharged they liberate an invisible gas-cloud.

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6.2.1 If particular loading, i.e., dry- or solid-loaded, or liquid-loaded charge, is required it should be so specified in the invitation for bids (see 3.1.2).

6.3 Service guarantee.- It is suggested that the puchasing officers include the following clause in the invitation for bids, and contracts:

Service guarantee.- The contractor guarantees to deliver to the Government, grenades which have been manufactured not more than 6 months prior to date of delivery; and furthermore, guarantees the grenades against defective material, workmanship, and performance for a period of 3 years from date of manufacture, provided they are stored in a cool, dry place. (The date of manufacture to be legibly marked on each grenade.) The contractor agrees to furnish, without cost to the Government, replacement of all grenades found to be defective during the guarantee period.

6.4 Questionnaire.- The invitation for bids should require bidders to completely fill out and submit with their bid one questionnarie sheet, page 12 of this specification, for each type, class, and composition grenade being offered.

NOTICE:- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Bids must be accompanied by the following questionnaire completely filled out by the bidder for each type, class and composition of grenade being offered:

1. Manufacturer's model or type of grenade _____.

2. Grenade over-all dimensions: Height _____ (inches) diameter _____ (inches).

3. Weight of loaded grenade: _____, _____.

4. Type of gas-agent loading and content of agent, or agents: (a) _____. (type of gas-agent: CN, DM, DA, EBA, etc.) (b) _____. (content of gas-agent or agents: each in gram

5. Time interval for discharge of grenade between throw and discharge: _____ (seconds).

6. Duration of grenade's discharge: _____ (seconds).

7. Manufacturer's additional performance and specification data, if any: __

_____

_____

_____

8. The grenades we propose to furnish deviate from the specification as follows: _____

_____

_____

9. It is agreed that in the event deviations are not listed or filed under paragraph 8 above, the right is reserved by the Government to demand full and complete compliance with the specification requirements.

Signed _____

Title _____

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